

Logarithmic Transformations in Linear Regression: Four Models and How to Interpret Them

Primary Reference: Benoit, K. (2011). Linear Regression Models with Logarithmic Transformations. Methodology Institute, London School of Economics.

I. Introduction and Context

1a. Why I Chose This Paper

I chose Benoit's paper because it matched the exact problem I had when I first picked this topic. I knew how to run a regression and report the result, but I was still unsure how to think about log transformations. If I saw a model with $\log(Y)$, $\log(X)$, or both, I usually could not explain clearly why that form was used instead of another one.

What stood out to me most in this paper was the 2×2 table on the first page. It breaks the problem into four cases: transform Y or not, and transform X or not. That sounds simple, but it helped me a lot. Before reading this paper, I did not really have a clean way to organize those models in my head. A longer textbook chapter might have been more detailed, but I felt I first needed a basic framework that I could actually remember and use.

1b. What the Paper Is About

At a basic level, this paper is about how to interpret regression coefficients when logarithms are used in the model. Benoit goes through four cases: no transformation, $\log(X)$ only, $\log(Y)$ only, and both variables logged. For each one, he shows what the coefficient β means once we move back from the log scale to ordinary units.

The part I found most useful is that the paper keeps bringing the interpretation back to something concrete. Once a logarithm appears in the model, the coefficient usually stops being a simple additive change and starts being about multiplicative or percentage change. The paper also explains why log transformations are often useful in the first place. They can shrink very large values, reduce right-skewness, and make a curved relationship look more linear. Benoit illustrates this with World Bank examples involving urbanization, per capita GNP, and infant mortality.

1c. Intended Audience and Field Context

This paper seems to be written for graduate students and applied researchers in the social sciences, especially people who use regression models in practice but may not feel fully comfortable interpreting logged variables. Although the author is a political scientist at the London School of Economics, the paper is really broader than political science. It fits into applied regression and quantitative methodology more generally.

What makes the paper useful is that the problem it discusses comes up all the time. Logged variables appear often in economics, political science, sociology, and other fields where variables like income, population, or mortality can vary a lot in scale. Because of that, it is easy to use a log transformation without being fully clear about what the coefficient means. I think this is why the paper is still helpful: it does not try to develop new theory, but instead explains a common practical problem in a way that is much easier to follow than many textbook treatments.

1d. How I Approached the Paper

I followed Keshav's three pass method, but I applied it to a short methods paper rather than a full research article. On the first pass, I read only the section titles, Table 1, and the bullet point summaries at the end of each section. That gave me the basic structure. I could see that the paper was organized around four models, and that each model had its own rule for interpreting the coefficient. This step helped more than I expected because once I understood the overall structure, the later algebra was much easier to follow.

On the second pass, I worked through the algebra more carefully and stopped whenever a step did not make sense right away. The linear-log section took the most effort. In that part, Benoit uses the fact that $\log(X) + 1 = \log(eX)$, which turns a one unit change in $\log(X)$ into a multiplicative change in X . I did not understand that immediately, so I went back to the log rules listed on the first page and checked the step for myself.

On the third pass, I tried to verify the main results on my own in R. I used simulated data that matched the reported coefficients for the linear-log, log-linear, and log-log examples. Then I checked whether my back transformed interpretations matched the paper's conclusions. This step was especially useful because it showed me that I was not just repeating formulas. It also revealed one mistake in my understanding, which I describe later in Section 2b.

II. Conceptual Engagement

2a. What the Paper Is Actually Doing, Statistically

At the core, this paper is about interpretation. Benoit is trying to show how we should interpret a regression coefficient when one or both variables are log transformed. The main statistical idea is not a new model. Instead, it is a way of moving between the log scale and the original scale without losing the meaning of the coefficient.

The key algebra behind this is simple. If we take the exponential of a logarithm, we return to the original value, and if we take the logarithm of an exponential term, we recover the original exponent. In this paper, those relationships make it possible to turn coefficients on the log scale into statements about changes in ordinary units or percentages.

The paper then organizes the four regression models into a 2×2 table based on whether *X* is logged, whether *Y* is logged, or whether both are logged. That table was one of the most helpful parts of the paper for me because it turned what first looked like several unrelated formulas into one simple structure.

	X (unlogged)	log(X)
Y (unlogged)	Linear: $Y = \alpha + \beta X$	Linear-log: $Y = \alpha + \beta \cdot \log(X)$
log(Y)	Log-linear: $\log(Y) = \alpha + \beta X$	Log-log: $\log(Y) = \alpha + \beta \cdot \log(X)$

Table 1. The four logarithmic transformation models (adapted from Benoit 2011, Table 1).

To make the four cases easier to compare, I created Figure 1 as a simple visual schematic of the model shapes on the original scale.

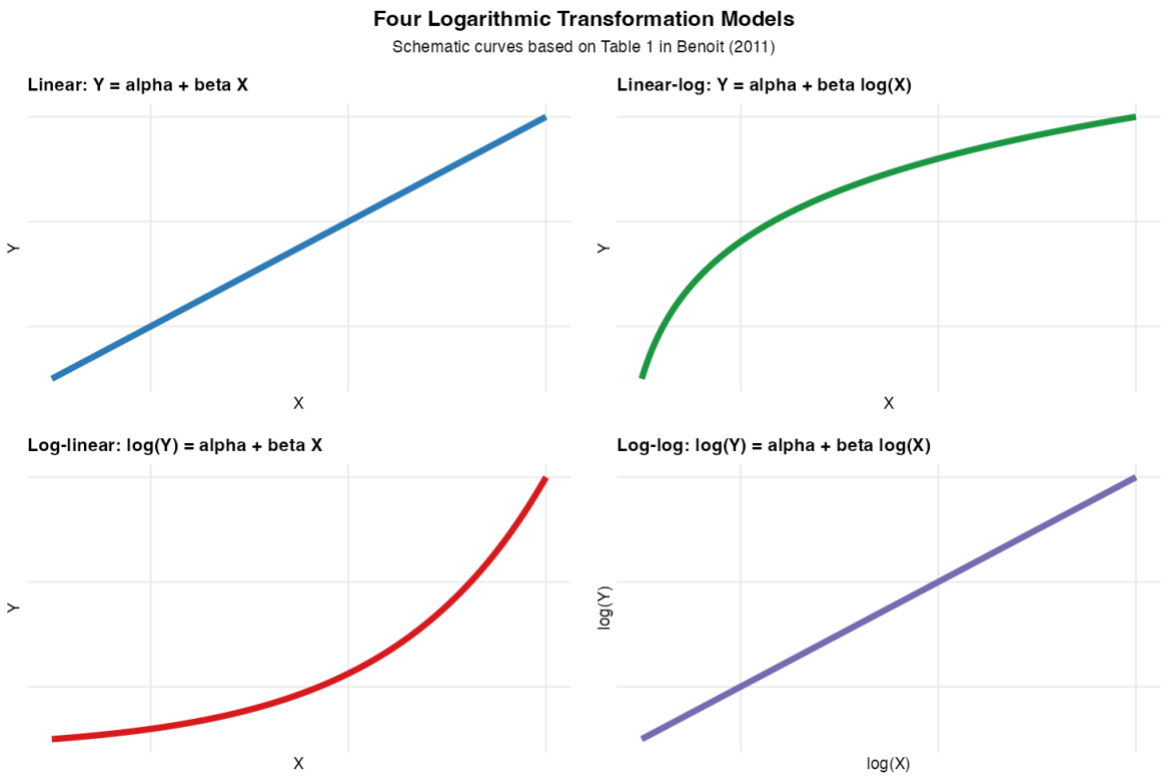


Figure 1. Visual schematic of the four logarithmic transformation models on the original scale.

The interpretation rules for the four models are summarized in Table 2. What helped me most was thinking about the models in a simpler way. The linear model is additive in both variables. The linear-log model treats proportional changes in X as producing additive changes in Y . The log-linear model treats additive changes in X as producing proportional changes in Y . The log-log model treats both X and Y proportionally. This way of thinking helped me move from memorizing formulas to understanding what kind of relationship each model is trying to capture.

Model	What β means (in plain terms)
Linear (no logs)	A one-unit increase in X increases Y by β units. (Additive, direct.)
Linear-log (log X only)	Multiplying X by e increases Y by β units. Equivalently: a 1% increase in X increases Y by $\beta/100$.
Log-linear (log Y only)	A one-unit increase in X multiplies Y by e^β . Equivalently: a 1-unit increase in X changes Y by approximately 100β percent.
Log-log (both logged)	A 1% increase in X changes Y by $\approx \beta$ percent. β is called the elasticity of Y with respect to X .

Table 2. Summary of coefficient interpretation across the four models.

The linear log case was the hardest one for me to understand. Benoit shows that a one unit increase in $\log(X)$ means multiplying X by e , not adding e to X . The key step is that adding 1 to $\log(X)$ gives $\log(X) + 1 = \log(X) + \log(e) = \log(eX)$. So the new value of X becomes eX , not $X + 1$.

What helped me here was realizing that the model is not describing equal additive changes in X . Instead, it is describing proportional changes in X that lead to additive changes in Y . That is why this model fits situations with diminishing returns. As X gets larger, the same proportional increase in X leads to the same additive change in Y .

Figure 2 shows this case using simulated data designed to match Benoit's reported regression of urbanization rate on per capita GNP.

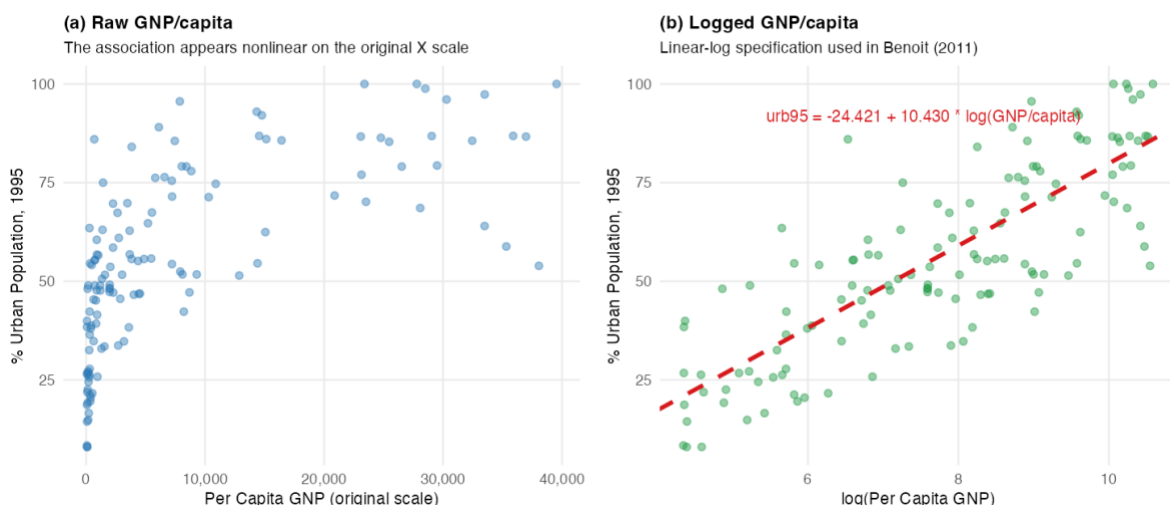


Figure 2. Raw and logged versions of per capita GNP in the linear log example. On the original scale, the relationship with urbanization appears nonlinear and more spread out. After logging GNP, the relationship becomes closer to linear.

Figure 3 helps show what changes in the log-linear case. On the log scale, the fitted relationship is a straight line. On the original scale, that same relationship becomes exponential. This is why it makes more sense to interpret the coefficient through the multiplicative factor e^β .

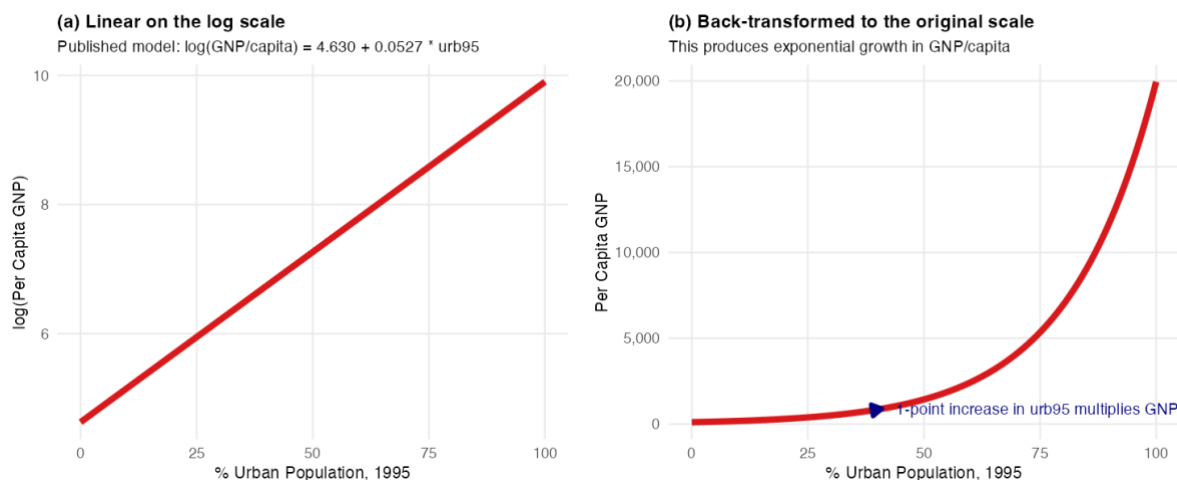


Figure 3. Log-linear model based on Benoit's reverse regression of urbanization on log per capita GNP, with $\beta = 0.0527$. Panel (a) shows the fitted relationship on the log scale. Panel (b) shows the same relationship on the original scale after back transformation. A one point increase in urb95 multiplies per capita GNP by $e^{0.0527} \approx 1.054$, or about 5.4 percent.

Figure 4 shows the log-log case, where infant mortality and GNP per capita are both logged. In this setting, β is an elasticity. Using Benoit's reported coefficient of $\beta = -0.4985$, a 10 percent increase in GNP per capita is associated with about a 4.6 percent decrease in infant mortality.

Log-log Model: Infant Mortality vs. GNP/capita

Published example from Benoit (2011), Section 3.4

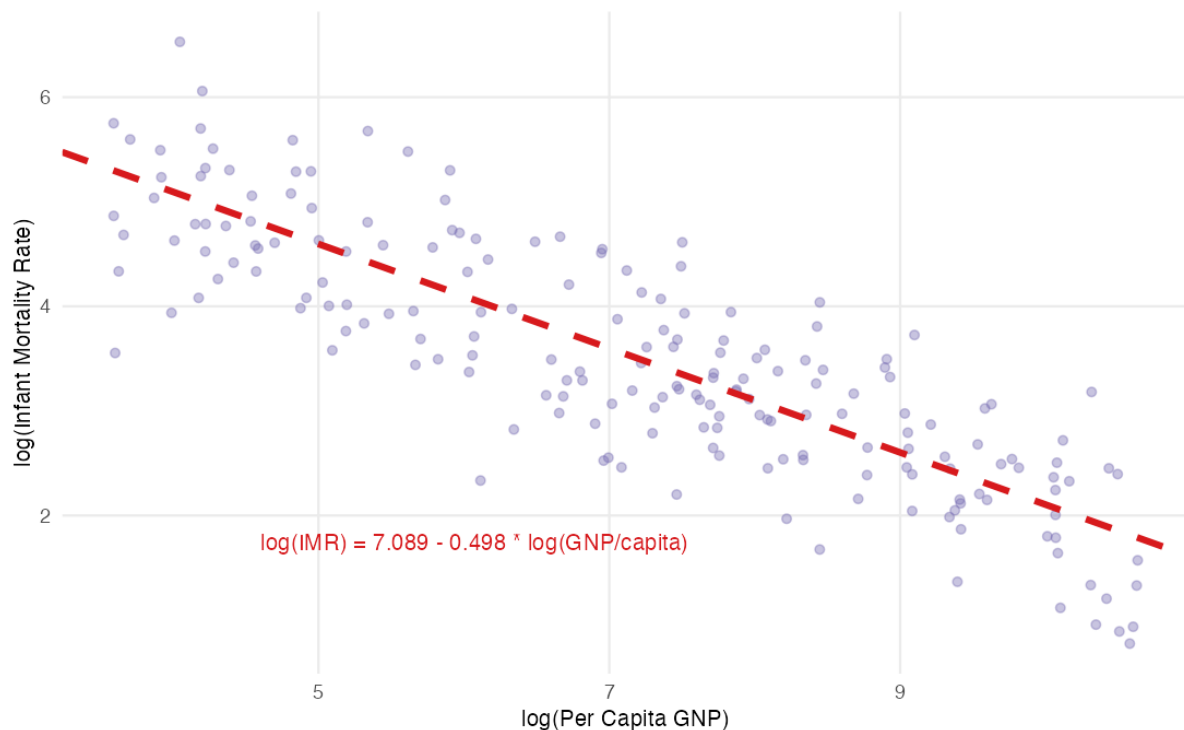


Figure 4. Log-log model of infant mortality and GNP per capita, with both variables on the log scale. The simulated data are designed to match Benoit's reported coefficient of $\beta = -0.4985$ and $R^2 \approx 0.678$.

2b. What I Understood vs. What Remained Unclear

What I understood most clearly after working through the paper:

- **The 2×2 taxonomy.** Organizing the four models by which variables are logged gave me a structure I did not have before. Now, when I look at a regression equation, I can usually tell which interpretation rule applies.
- **The linear-log interpretation.** Once I worked through the algebra, the key point became much clearer to me. If $\log(X)$ increases by 1, then X is multiplied by e . From there, the rule that $\beta/100$ gives the approximate change in Y for a 1 percent increase in X started to feel intuitive instead of something I was only memorizing.
- **The log-log interpretation.** This part clicked for me because I had seen the idea of elasticity before in economics. If both variables are logged, then the coefficient tells us the percentage change in Y associated with a percentage change in X . In Benoit's example, a negative coefficient means that higher GNP is associated with lower infant mortality in proportional terms.
- **Why logging helps with right skew.** I understood this much better after looking at the

examples. The log function compresses very large values and spreads out smaller ones. That makes the variable less skewed and often makes the relationship look closer to linear.

What still felt unclear to me:

- **The approximation $e^\beta \approx 1 + \beta$ for small β .** The paper uses this in the log linear section to justify reading 100β as an approximate percentage change in Y for a one unit change in X . I understand where this comes from, since it follows from the Taylor expansion of e^x around $x = 0$. What I am still unsure about is when β is small enough for this approximation to be reliable. In Benoit's example, $\beta = 0.0527$ gives 5.27 percent by approximation and about 5.4 percent using the exact expression. Those are close, but not identical.
- **The general formula for a p percent change in the linear-log model.** Benoit gives the result that the expected change in Y for a p percent increase in X is $\beta \log((100 + p)/100)$. I verified this numerically, but I could not derive it cleanly the first time I read the paper. I had to work backward from the formula before the derivation made sense.
- **How to decide which model to use.** The paper is mainly about interpretation, not model selection. I understand what each choice means for the coefficient, but I still do not have a systematic way to decide whether to log transform X , Y , both, or neither in a real dataset. In particular, I am not yet confident that I could look at a real dataset and decide whether a log transformation is appropriate without relying on trial and error.

2c. A Question the Paper Leaves Unanswered for Me

The paper focuses on how to interpret coefficients after a transformation has already been chosen. What it does not really address, at least for me, is a more statistical question: if we choose a transformation after looking at the data, what does that do to the validity of later inference, especially standard errors and hypothesis tests?

More specifically, in a log-linear model, the residuals are evaluated on the log scale. If those residuals are approximately normal and homoscedastic, then OLS is valid on that scale. But in Benoit's Section 4, the motivation for taking $\log(X)$ is that the raw GNP variable is right skewed and shows heteroscedasticity. That means the transformation is chosen after inspecting the data. I do not know how much that affects later inference. In particular, I wonder whether that kind of data based transformation choice could inflate the Type I error rate for later tests on β .

My guess is that the effect is probably small in this example, because the transformation is being used to address a distributional problem rather than to maximize a test statistic. Still, I am not sure, and the paper does not discuss it. This question feels important to me because it connects the paper to broader issues of model selection, researcher degrees of freedom, and what counts as valid inference after analytic choices have already been made.

III. Self-Directed Learning

3a. Outside Resources I Consulted

Resource	How It Helped
StatQuest (YouTube): “Linear Regression” series	Helped me build a visual intuition for what a regression coefficient means before I moved to logged models.
Khan Academy: “Logarithms” unit	Helped me review the log rules I needed in order to follow Benoit’s derivations.
R (ggplot2, dplyr)	I used R to reproduce the four model forms with simulated data and check whether my interpretations matched the implied fitted values.
Kutner et al., Chapter 3	Helped me see that model interpretation and model selection are different problems, and pointed me toward Box Cox and residual diagnostics.

3b. A Concept I Taught Myself: Elasticity

Before reading this paper, I had seen the word “elasticity” in economics, but I had never connected it to regression. Benoit mentions in Section 3.4 that in the log-log model, the coefficient β is interpreted as an elasticity. I had to look this up because I understood the word only loosely and did not know what it meant in a statistical model.

What I learned is that elasticity measures the percentage change in Y relative to the percentage change in X . In symbols,

$$\varepsilon = \frac{\Delta Y/Y}{\Delta X/X} = \frac{dY}{dX} \cdot \frac{X}{Y}.$$

In the log-log model,

$$\log(Y) = \alpha + \beta \log(X).$$

If we differentiate both sides with respect to $\log(X)$, then $d\log(Y)/d\log(X) = \beta$. Since $d\log(Y)/d\log(X)$ is the same as $(dY/Y)/(dX/X)$, the coefficient β is exactly the elasticity of Y with respect to X .

This mattered for me because it changed the way I thought about the model. The log-log model is not only a convenient mathematical form. It also represents a specific way of describing the

relationship between X and Y . It says that the relationship is better understood in proportional terms rather than in absolute units. In Benoit's example, a coefficient of $\beta = -0.4985$ means that a 10 percent increase in GNP is associated with about a 4.6 percent decrease in infant mortality. That is a scale free interpretation, which is one reason the model makes sense for cross country comparisons.

3c. A Plan for Further Study

The biggest gap this paper left for me is the model selection question. Given a real dataset, how do I decide which of the four transformation cases is appropriate? Benoit's paper assumes that this choice has already been made and focuses on interpretation. Because of that, my next step would be to read Kutner et al., Chapter 3, "Diagnostics and Remedial Measures," more carefully.

That chapter would help with the part that Benoit does not really cover. It discusses residual plots, the Box Cox likelihood approach for choosing a power transformation, and formal tests for heteroscedasticity such as Breusch Pagan and Levene. Together, Benoit and Kutner would give me something closer to a full workflow. Kutner would help me think about how to choose a transformation, and Benoit would help me interpret the result once that choice is made.

A second step would be to apply both papers to a real dataset. A World Bank dataset would be a natural place to start, since Benoit's examples also use cross national economic data. I think actually working through the whole process would help me most. That would mean looking at the raw variables, deciding whether a transformation is appropriate, fitting the model, and then checking whether I can interpret the coefficient correctly. Right now, these are still ideas I can explain more easily than ideas I can use with confidence.

References

- [1] Benoit, K. (2011). *Linear Regression Models with Logarithmic Transformations*. Methodology Institute, London School of Economics. Primary reference.
- [2] Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied Linear Statistical Models* (5th ed.). McGraw Hill. Consulted for model selection and diagnostics, especially Chapter 3, "Diagnostics and Remedial Measures."
- [3] StatQuest with Josh Starmer. *Linear Regression and Linear Models*. YouTube.
- [4] Khan Academy. *Logarithms*. Reviewed for log properties used in the paper.

Note on AI Use

I used an AI assistant for formatting help and for checking the clarity of some written explanations. The choice of paper, the statistical interpretation, the reflection on what I understood and what remained unclear, and the plan for further study are my own.